COMPACT MAPS INTO THE TOWER FACTOR THROUGH A FINITE SUBFAMILY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11440-11480, following the statement at 11431-11438.

Editorial note. Author typo kept verbatim: the third induction index is written E"/I' where E"'/I"' is meant; no mathematical content altered.

Editorial note. No omissions: the mismatch between the constructed and the given map is handled by lifting the difference to X_{n-1} and applying the induction hypothesis again.

Allowed standard prerequisites (editorial).

- The tower of lemma-write-as-colimit. lemma-write-as-colimit (carded L26187 in this batch); its triangles Y_n -> X_n -> X_{n+1} -> Y_n[1] are the induction data. Supplies the triangles the induction climbs.
- Triangulated axioms. TR1 and TR3 of the same chapter. Build the maps of triangles in both induction steps and lift the difference to X_{n-1}.

Source reference. Original lemma 09SN: sources/stacks-derived/pinned-original-source.tex, source lines 11386–11386. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With assumptions and notation as in Lemma 09SN. If C is a compact object and $C \to X_n$ is a morphism, then there is a factorization $C \to E \to X_n$ where E is an object of $\langle E_{i_1} \oplus \ldots \oplus E_{i_t} \rangle$ for some $i_1, \ldots, i_t \in I$.*

Proof. We prove this by induction on n. The base case $n = 1$ is clear. If $n > 1$ consider the composition $C \to X_n \to Y_{n-1}[1]$. This can be factored through some $E'[1] \to Y_{n-1}[1]$ where E' is a finite direct sum of shifts of the E_i. Let $I' \subset I$ be the finite set of indices that occur in this direct sum. Thus we obtain

$$\begin{array}{ccccc} E' & \longrightarrow & C' & \longrightarrow & C & \longrightarrow & E'[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ Y_{n-1} & \longrightarrow & X_{n-1} & \longrightarrow & X_n & \longrightarrow & Y_{n-1}[1] \end{array}$$

By induction the morphism $C' \to X_{n-1}$ factors through $E'' \to X_{n-1}$ with E'' an object of $\langle \bigoplus_{i \in I''} E_i \rangle$ for some finite subset $I'' \subset I$. Choose a distinguished triangle

$$E' \to E'' \to E \to E'[1]$$

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then E is an object of $\langle \bigoplus_{i \in I' \cup I''} E_i \rangle$. By construction and the axioms of a triangulated category we can choose morphisms $C \to E$ and a morphism $E \to X_n$ fitting into morphisms of triangles $(E', C', C) \to (E', E'', E)$ and $(E', E'', E) \to (Y_{n-1}, X_{n-1}, X_n)$. The composition $C \to E \to X_n$ may not equal the given morphism $C \to X_n$, but the compositions into Y_{n-1} are equal. Let $C \to X_{n-1}$ be a morphism that lifts the difference. By induction assumption we can factor this through a morphism $E''' \to X_{n-1}$ with E'' an object of $\langle \bigoplus_{i \in I'''} E_i \rangle$ for some finite subset $I' \subset I$. Thus we see that we get a solution on considering $E \oplus E''' \to X_n$ because $E \oplus E'''$ is an object of $\langle \bigoplus_{i \in I' \cup I'' \cup I'''} E_i \rangle$. $\square$